

Shahid Karimi

Senior Backend Architect | Distributed Systems & High-Throughput Data Platforms | karimi.shahid@gmail.com | [+49 163 4733581](tel:+491634733581) | linkedin.com/in/shahidkarimi | Germany

About me

I build backend systems that stay fast and reliable under real load. Over 12+ years I have engineered high-throughput data pipelines and distributed crawling infrastructure at Zyte, scaled backend services for the Cloudways cloud-hosting platform, and re-architected core request paths as Software Architect at Marengo Knoll (HelloGram.AI) — cutting response latency by 80% through systematic profiling and PostgreSQL optimisation. Deep foundations in Python and PHP/Laravel services, PostgreSQL/MySQL/Redis, message queues, Docker, AWS and CI/CD, with a strong bias toward clean architecture, testability and operational ownership.

Selected Impact

- **80% latency reduction** — re-architected core request paths and optimised PostgreSQL under production load at Marengo Knoll.
- **High-volume data delivery** — built and operated distributed crawling and extraction pipelines at Zyte powering dependable data feeds for major retailers.
- **Platform scale** — developed and maintained highly available backend services and microservices for the Cloudways cloud-hosting platform.
- **Legacy modernisation** — repeatedly turned untested legacy codebases into maintainable, testable systems using characterisation tests and the strangler-fig pattern.

Work Experience

Software Architect, Marengo Knoll, Inc. (HelloGram.AI)

March 2023 - September 2025

- Software Architect for HelloGram.AI, a real-time conversational AI platform combining LLMs, voice recognition and voice-cloning technologies.
- Designed and scaled cloud-native backend services and APIs for data-intensive, low-latency production workloads.
- Re-architected core request paths and optimised PostgreSQL under load, cutting response latency by 80%.
- Led integration of AI, LLM and voice-processing pipelines with reliable async processing.

- Owned production investigation, deployments and technical documentation, strengthening release confidence.

Independent Software Consultant, Self-Employed

October 2022 - February 2023

- Delivered Python and PHP/Laravel backend services, REST APIs and full-stack applications for international clients.
- Designed scalable deployments on AWS, Azure and DigitalOcean with Docker and CI/CD pipelines.
- Improved performance, reliability and code quality through pragmatic architecture and testing.

Python Developer | Crawl Engineer, Zyte Data Ltd.

July 2021 - September 2022

- Engineered high-performance distributed web crawlers powering data pipelines for major retailers.
- Developed and maintained production-grade Scrapy spiders extracting structured data from complex websites with high reliability.
- Modernised legacy codebases to improve testability, maintainability and confidence in production changes.
- Solved complex anti-bot, extraction and website-change problems through systematic root-cause analysis.

Software Engineer, Cloudways

March 2018 - February 2020

- Developed and maintained highly available backend services and microservices for a managed cloud-hosting platform.
- Architected microservices for server provisioning and orchestration, enhancing system reliability and efficiency.
- Improved production resilience using MySQL, Redis, AWS, Linux, queues and deployment automation.
- Refactored large codebases to reduce technical debt and improve maintainability across platform delivery.

Software Engineer, Thumbsol (now Arrivy)

March 2013 - July 2017

- Designed and deployed web services with Laravel and Yii, delivering scalable and efficient backend solutions.
- Optimised databases for high-traffic web applications, enhancing performance and reliability.
- Built search, real-time and responsive product features across PHP, JavaScript and Node.js.
- Owned testing, debugging and production releases, improving long-term stability.

Education

M.Sc. Data Science, Freie Universität Berlin

Enrolled — studies begin October 2026

Bachelor of Computer Science, Karakoram International University

2005 - 2011 — ZAB-recognised in Germany

Skills

- Backend: Python, PHP 7/8, Laravel, Symfony, FastAPI, REST APIs, Microservices
- Data & Storage: PostgreSQL, MySQL, Redis, Elasticsearch, ETL, Data Pipelines
- Distributed Systems: Queues, Async Processing, Caching, Scrapy, High-Throughput Crawling
- Cloud & Delivery: AWS, Azure, Docker, Linux, Nginx, CI/CD
- Frontend: Vue.js, React, JavaScript/TypeScript

Certifications

- Google Data Analytics Professional Certificate - 2022

Projects

Hellogram.ai

Real-time conversational AI platform enabling lifelike voice interactions with digital personas, built on LLMs, voice recognition and voice-cloning pipelines.

Roomosys.com

Hotel management system built with Python 3.12, ReactJS, Supabase, Golang and AWS.

Bewhy.app

Healthcare SaaS built with PHP/Laravel, PostgreSQL, Vue.js and Microsoft Azure.

Zoomkar.com

Real-estate and property management platform built with Python, ReactJS, AWS and Docker.

Monolith to Microservices Migration

Converted a high-traffic monolith into a scalable microservice architecture with zero downtime.

Languages

- English: Fluent
- German: A2, actively learning
- Urdu: Fluent
- Hindi: Fluent
- Balti (Tibetan): Native

Interests

- Hiking
- Technology
- Human History